

■ PySpark 30-Day Practice Plan

Complete Roadmap — Beginner to Job-Ready Data Engineer

Platform: Databricks Community Edition | Language: Python / PySpark
Stack: Apache Spark 4.1 · Delta Lake · Azure · GitHub
Includes: Data Modeling · Star Schema · SCD · Auto Loader · Workflows

■ Learner: Pradnya Nitin
■ Goal: Data Engineering role in India → Europe

■ Complete 30-Day Overview

Week	Days	Focus Area	Key Topics
Week 1	Days 1–7	PySpark Foundations	SparkSession, DataFrame, filter, groupBy, Nulls, Read/Write
Week 2	Days 8–14	Transformations + SQL + Window Functions	Joins, Spark SQL, Window Functions, UDFs, Set Ops, Mini Project
Week 3	Days 15–21	Data Modeling + Databricks Core	OLTP/OLAP, Star/Snowflake Schema, SCD, Delta Lake, Auto Loader, Spa
Week 4	Days 22–30	Production + Interview Prep	Workflows, Unity Catalog, Streaming, Azure, End-to-End Project, Mock Inte

Legend:

■ Completed

■ New / Added topic

■ In Progress

■ Upcoming

■ WEEK 1 — PySpark Foundations (Days 1–7)

Day	Topic	What You Learn / Build	Status
Day 1	SparkSession + DataFrame Basics	Create SparkSession, createDataFrame, show(), printSchema(), describe(), count(), columns, dtypes	■
Day 2	select, filter, where, col()	select(), filter(), where(), col(), alias(), isin(), between(), lit(), & ~ operators	■
Day 3	withColumn, cast, drop, rename	withColumn(), cast(), IntegerType, when().otherwise(), withColumnRenamed(), drop(), to_date(), year()	■
Day 4	groupBy + Aggregations	groupBy(), agg(), count, sum, avg, max, min, round, countDistinct, collect_list, collect_set, HAVING	■
Day 5	orderBy, distinct, dropDuplicates	orderBy(), sort(), desc(), asc(), limit(), distinct(), dropDuplicates(), dropDuplicates([cols])	■
Day 6	Null Handling	isNull(), isNotNull(), dropna(), fillna(), coalesce(), when().otherwise() for nulls, null audit	■
Day 7	Read/Write + Delta Tables	createDataFrame with schema, saveAsTable(), read.table(), partitionBy(), DROP TABLE IF EXISTS, overwriteSchema	■

■ WEEK 2 — Transformations + SQL + Window Functions (Days 8–14)

Day	Topic	What You Learn / Build	Status
Day 8	Joins — All Types	inner, left, right, full, left_anti, left_semi, multi-col join, broadcast join, alias for ambiguous cols	■
Day 9	Spark SQL + Temp Views	createOrReplaceTempView(), spark.sql(), SELECT/WHERE/GROUP BY/HAVING/CASE WHEN/Subqueries, %sql magic	■
Day 10	Window Functions — Ranking	Window.partitionBy().orderBy(), row_number(), rank(), dense_rank(), ntile(), global vs partitioned rank	■
Day 11	Window Functions — Lag/Lead	lag(), lead(), running total, moving average, rowsBetween(), unboundedPreceding, currentRow, first(), last()	■
Day 12	UDFs	Python UDF, udf() registration, multi-col UDF, spark.udf.register(), Pandas UDF (@pandas_udf), built-in vs UDF	■
Day 13	Set Operations	union(), unionByName(), distinct after union, intersect(), exceptAll(), data reconciliation, multi-source pipeline	■
Day 14	Week 2 Mini Project	Sales Analytics Pipeline: Q1+Q2 union → clean → enrich (join) → rank (window) → trends (lag) → SQL → Delta	■

■ WEEK 3 — Data Modeling + Databricks Core (Days 15–21)

Day	Topic	What You Learn / Build	Status
Day 15	OLTP vs OLAP + Lazy Eval + DAG	OLTP vs OLAP differences, transformations vs actions, lazy execution, DAG stages, shuffle triggers ■	■
Day 16	Fact + Dimension Tables + Star Schema	Fact vs dimension, star schema design, build retail data warehouse in Delta Lake ■	■
Day 17	Snowflake Schema + Surrogate Keys	Snowflake vs star, normalised dimensions, monotonically_increasing_id(), row_number() as surrogate key ■	■
Day 18	SCD Type 1 + SCD Type 2	Slowly Changing Dimensions, Type 1 overwrite, Type 2 new row with is_current/start_date/end_date via MERGE ■	■
Day 19	Delta Lake — MERGE + Time Travel	DeltaTable.merge(), whenMatchedUpdate, whenNotMatchedInsert, versionAsOf, timestampAsOf, VACUUM, OPTIMIZE ■	■
Day 20	Auto Loader + Shuffle	cloudFiles format, Auto Loader incremental ingestion, shuffle deep dive, broadcast, AQE, coalesce vs repartition ■	■
Day 21	Spark UI + Full Optimization	Read DAG in Spark UI, identify slow stages, partition tuning, caching strategy, executor memory config ■	■

■ WEEK 4 — Production + Interview Prep (Days 22–30)

Day	Topic	What You Learn / Build	Status
Day 22	Medallion Architecture	Bronze/Silver/Gold layers, design principles, implement full medallion pipeline with Delta tables ■	■
Day 23	Databricks Workflows + Job Scheduling	Create multi-task workflow, task dependencies, Cron scheduling, retries, email alerts, parameters ■	■
Day 24	Unity Catalog + Data Governance	Catalogs, schemas, tables hierarchy, grants/permissions, data lineage, row-level security basics ■	■
Day 25	Structured Streaming	readStream, writeStream, trigger options, checkpointing, watermarking, streaming aggregations ■	■
Day 26	Azure + Databricks + ADLS Gen2	Mount ADLS, service principal auth, read/write from Azure storage, Azure Data Factory integration overview ■	■
Day 27	End-to-End Project — Day 1	Design architecture, create Bronze layer, ingest raw data, data quality checks, document design decisions ■	■
Day 28	End-to-End Project — Day 2	Build Silver layer (clean + validate), build Gold layer (aggregations + star schema), window functions ■	■
Day 29	End-to-End Project — Day 3	SQL dashboard queries, Workflow orchestration, documentation, GitHub README, project presentation ■	■
Day 30	Mock Interview + Resume + GitHub	SQL round, PySpark round, system design round, resume review, GitHub portfolio polish, next steps ■	■

Certifications to Target

Certification	Cost	When to Attempt	Value
Databricks Associate Developer for Apache Spark	\$200 (~\$16,500)	After Day 25	■■■■■
Azure Data Engineer Associate (DP-203)	\$165 (~\$13,600)	After Day 30	■■■■■
Azure Fundamentals (AZ-900)	\$165 (~\$13,600)	After Day 26	■■■
Google Professional Data Engineer	\$200 (~\$16,500)	3-6 months later	■■■■

Interview Topics Covered

Topic	Day Covered	Interview Question
PySpark DataFrames	Days 1–7	Filter/transform/aggregate data using PySpark
Joins + Broadcast	Day 8	How do you optimise a large table join?
Spark SQL	Day 9	Write SQL to find top N per group
Window Functions	Days 10–11	Rank employees per department by salary
Lazy Evaluation	Day 15	When does Spark actually execute code?
Shuffle	Day 20	How do you reduce shuffle in Spark?
Star Schema	Day 16	Design schema for e-commerce data warehouse
Snowflake Schema	Day 17	Star vs Snowflake — when to use which?
Fact vs Dimension	Day 16	What is the difference between fact and dimension?
SCD Type 1 + 2	Day 18	How do you handle slowly changing dimensions?
Surrogate Keys	Day 17	What is a surrogate key and why use it?
OLTP vs OLAP	Day 15	Explain OLTP vs OLAP with examples
Delta Lake MERGE	Day 19	How do you upsert data in Delta Lake?
Time Travel	Day 19	How do you query historical data in Delta?
Auto Loader	Day 20	How do you ingest streaming files incrementally?
Medallion Architecture	Day 22	Explain Bronze/Silver/Gold architecture
Databricks Workflows	Day 23	How do you schedule and orchestrate pipelines?
Unity Catalog	Day 24	How do you manage data governance in Databricks?
Structured Streaming	Day 25	Explain your experience with streaming pipelines
Azure + ADLS	Day 26	How do you connect Databricks to Azure storage?

■ Career Roadman — India to Europe

Phase	Timeline	Action Items	Goal
Phase 1 Build Skills	Now (0-3 months)	Complete 30-day plan, Databricks cert, Azure DP-203, GitHub portfolio with 30 projects	3 months skills
Phase 2 India Job	3-6 months	Apply to TCS/Infosys/Wipro/Capgemini/startups, target 6-12 LPA DE role	1-2 years experience
Phase 3 Europe Apply	1-2 years later	Apply via LinkedIn/Indeed.de, target Germany/Netherlands/Ireland, negotiate relocation package	EU relocation package
Phase 4 Europe	2-3 years	Settle in Germany/Netherlands, salary EUR 55,000-85,000, apply for PR after 5 years	Permanent residency

■ **Keep pushing — every day of practice brings you closer to your Europe dream!**

GitHub: github.com/PradnyaNitin/pyspark-30-days

Stack: PySpark · Databricks · Delta Lake · Azure · GitHub